Introduction and research questions

The California housing market is very dynamic and diverse. There have been lots of economically significant areas and is an economical powerhouse for the real estate industry in the United States. Housing prices across the state vary vastly due to many factors including location, rooms, income levels, population density, neighborhoods, and distance from the coastline. If we are able to understand the factors that influence housing prices, it can provide valuable information for buyers, investors, policymakers/politicians , and more who are dealing with the housing industry.

The goal of this project is to apply different data mining techniques to predict median housing values using a publicly available California housing dataset from 1990. This project aims to answer these two research questions which guided our work.

1. How accurately can housing prices in California be predicted using demographic and geographic features?
2. Which variables have the greatest influence on housing prices?

To answer these questions, we use KNIME Analytics Platform to build a complete data mining pipeline to explore, process, engineer, and model the housing market.

Dataset Description/Method

The dataset used contained 20,640 records, each representing a neighborhood of homes in a California area which are separated by distance from the coast. This dataset included many different variables, most notable were geographic location and housing related variables used to predict housing prices. Our target value was median house value, which represents the median market value of homes in each district (where each district is represented by 1 data point).

The predictor variables included:

- Longitude
- Latitude
- Housing median age
- Total rooms
- Total bedrooms
- Population
- Households
- Median income
- Ocean proximity classification

Ocean proximity was represented by five categories:

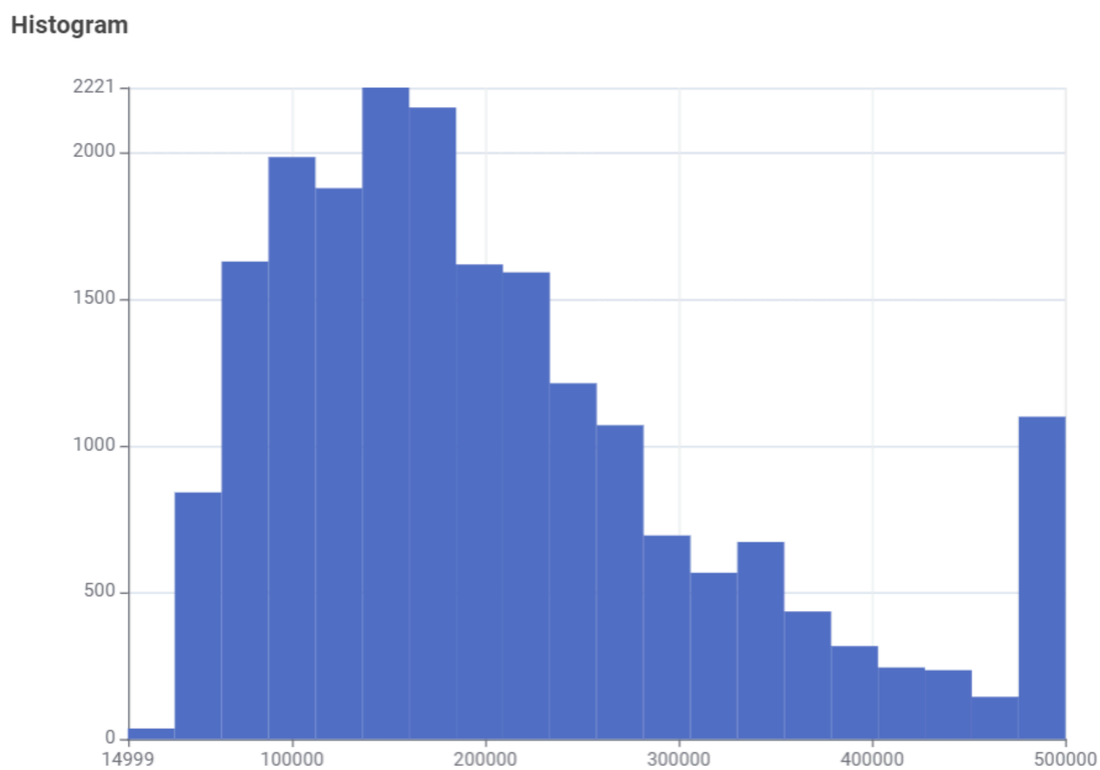
- NEAR BAY
- <1H OCEAN
- INLAND

- NEAR OCEAN
- ISLAND

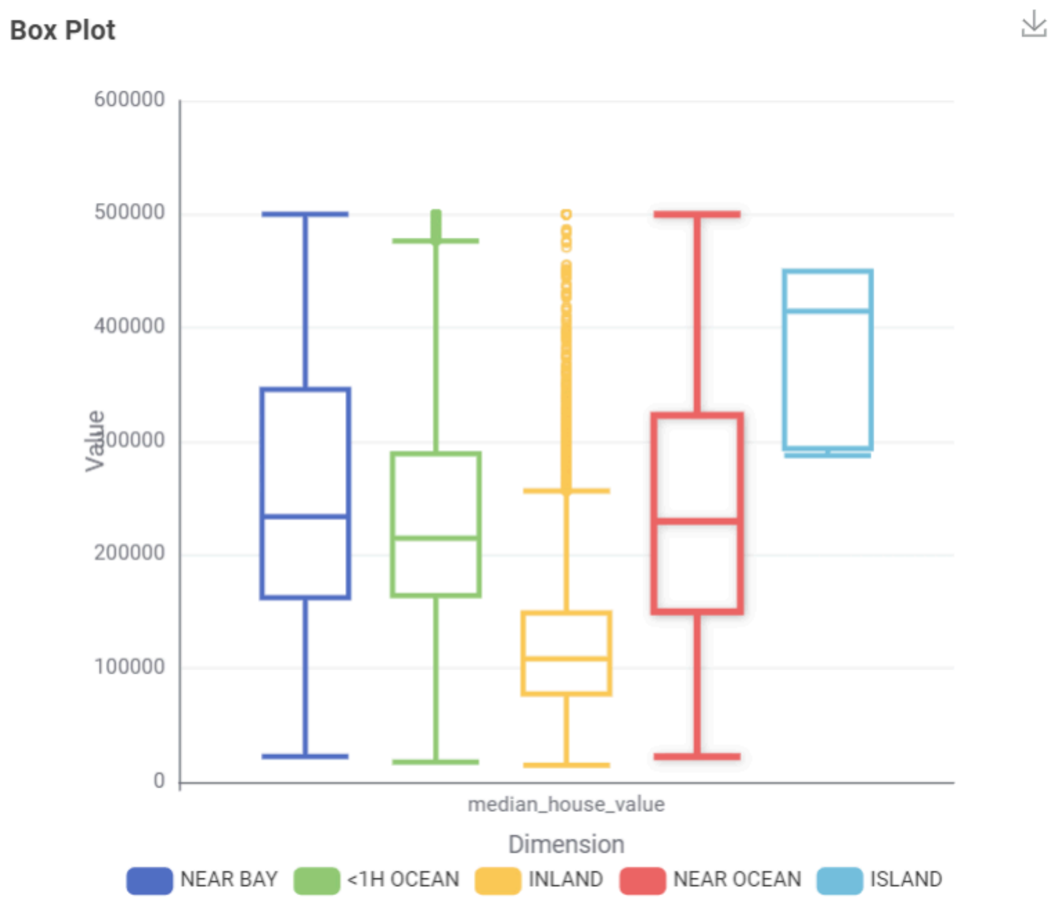
These variables were selected because they represent both economic and geographic factors that may influence property values.

Simple Data Analysis

Before major data analysis happened and predictive models were built, we did some exploratory data analysis to better understand the structure of the dataset to see obvious patterns, outliers, and potential issues.



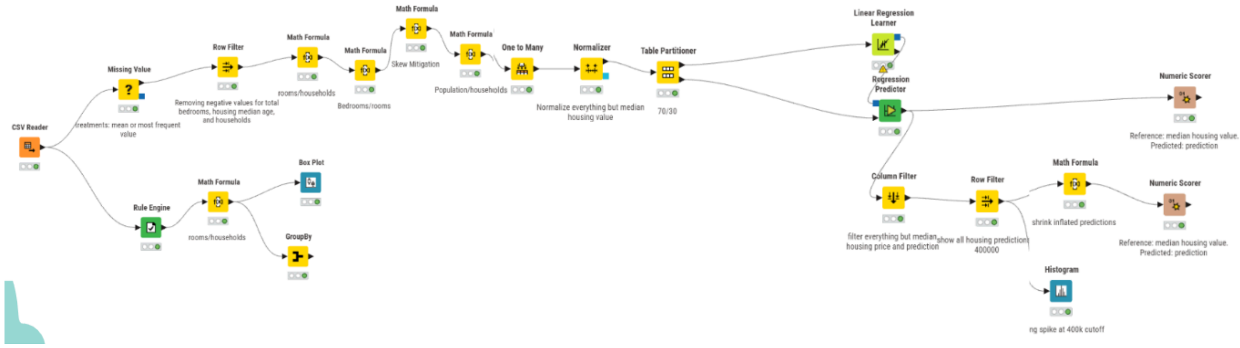
The histogram above was created in KNIME and shows the median housing values regardless of any variable. The y-axis represents the number of datapoints, and the x-axis represents the price of the homes up to $500,000, meaning that the data is capped at that amount within the entire dataset. This histogram also clearly shows right skewed data, with most of the data being largely concentrated between $100,000 and $200,000.



Further analysis using a box plot shows significant differences in housing prices across the region. Value on the y-axis represents money in USD and the x-axis represents median housing value. The colors represent a different geographic region. This data gives more insight into the housing market because island properties are typically on the higher end of pricing, while inland properties are largely the cheapest but have so many outliers that an inland property still might be more expensive than an island property. The bay area, <1H Ocean area, and the Near Ocean area typically have similar housing cultures in terms of pricing. This box plot alone suggests that geographic location is a highly important predictor of how housing is typically affected by region.

Here we learn 2 important things, the data is capped at $500,000, and geographic region paints a picture of how housing prices change by location but may not be an absolute predictor.

Data preprocessing



The figure above (while small) shows our entire finalized KNIME workflow, in this section, we will only talk about data preprocessing. This was our preprocessing workflow thoughts

1. Missing value treatment
   a. This node was quickly added as many data points (typically within housing related variables) had missing values. To combat the missing data issue, all numerical missing values were replaced by the mean.
2. One to many (one hot encoding)

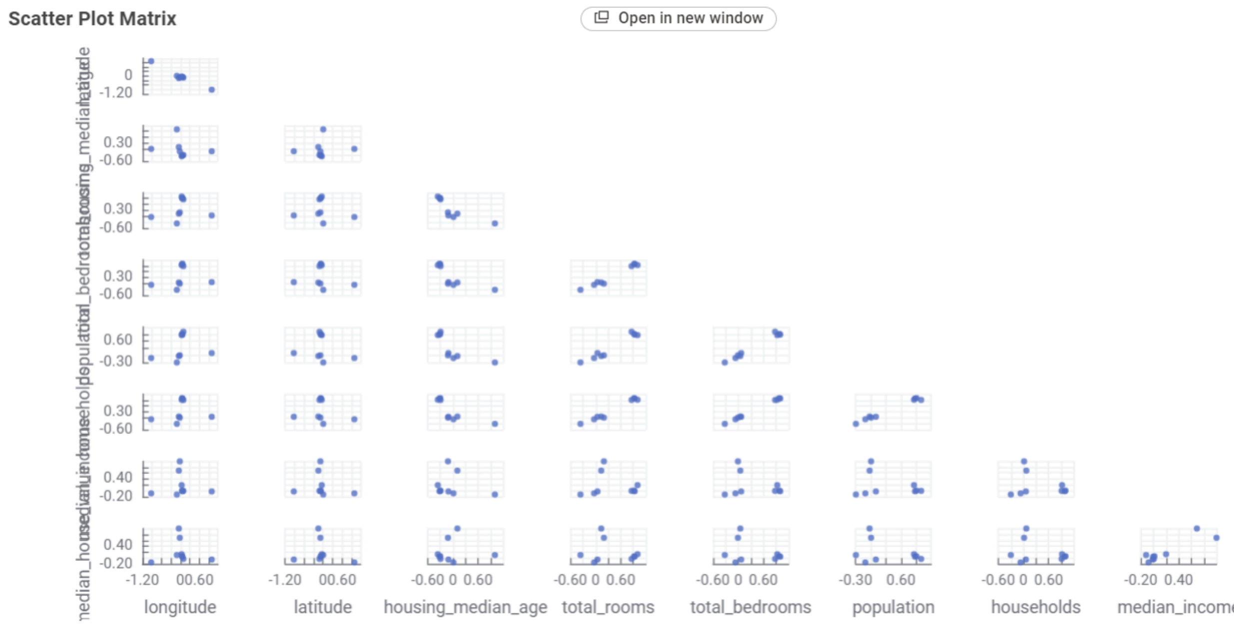
   a. This node was created to split the data up into many attributes to give a binary value. For example, instead of the data saying “near bay” a new attribute (a column) was created called “near bay” and then it would have a binary value of 1 or 0. 1 represents that the housing is near the bay area, 0 is the opposite. This allowed us to better understand the data in where it is geographically.
3. Feature Engineering

   a. Created using the 2 math formula nodes. We created relationships between variables to better understand how one affects another. We created population per household, and bedroom to room relationships. These created features helped improve interpretability and predictive performance … We could now group 2 variables together to see how they affected each other.

4. Normalization
   a. We made sure to normalize the variables before modeling to ensure that variables measured on different numerical scales were each taken into consideration fairly when using different modeling techniques.

Now that the data has been preprocessed, we could now start feature selection and modeling.

Feature Selection



Basic analysis and analysis from the preprocessing data showed that total rooms and total bedrooms typically have a correlation (it is hard to see but this scatter plot matrix helps support this analysis). Being able to see this type of correlation and analysis allowed for us to remove redundant features and helped simplify the model while reducing noise and improving interpretability.

The final selected features were:

1. Population
2. Geographic location
3. Median income
4. Households
5. Total rooms

These features were assessed to give us the best picture of how housing prices are shaped in California (1990).

Modeling Process

After preprocessing and feature selection were completed, the finalized dataset was divided into training and testing subsets using a 70/30 partition. Seventy percent of the data was used to train the predictive model, while the remaining thirty percent was reserved for testing and evaluating model performance unseen data. This split was chosen because it provides enough data for model training while still preserving a substantial portion for reliable performance evaluation.

A linear regression model was selected as our primary predictive modeling technique. Linear regression was chosen because the target variable, median housing value, is continuous and because the model offers strong interpretability, allowing us to examine how each predictor contributes to housing prices. Additionally, linear regression serves as a useful baseline model for structured tabular data and provides coefficients that help answer the research question regarding which variables most strongly influence housing values.

The model was implemented into KNIME using a workflow that connected the preprocessing pipeline directly into the regression learner and predictor nodes. The selected features (population, geographic location, median income, households, and total rooms) were fed into the linear regression learner node to train the model. Predictions were then generated on the testing subset using the regression predictor node, and model performance was evaluated using multiple regression metrics.

This workflow allowed for a reproducible end-to-end machine learning pipeline, from raw data preprocessing through final model evaluation.

Results

The final linear regression model performed strongly in predicting California housing prices, achieving the following evaluation metrics:

- $R^2 = 0.914$
- Mean Absolute Error (MAE) = $26,604
- Mean Absolute Percentage Error (MAPE) = 19%

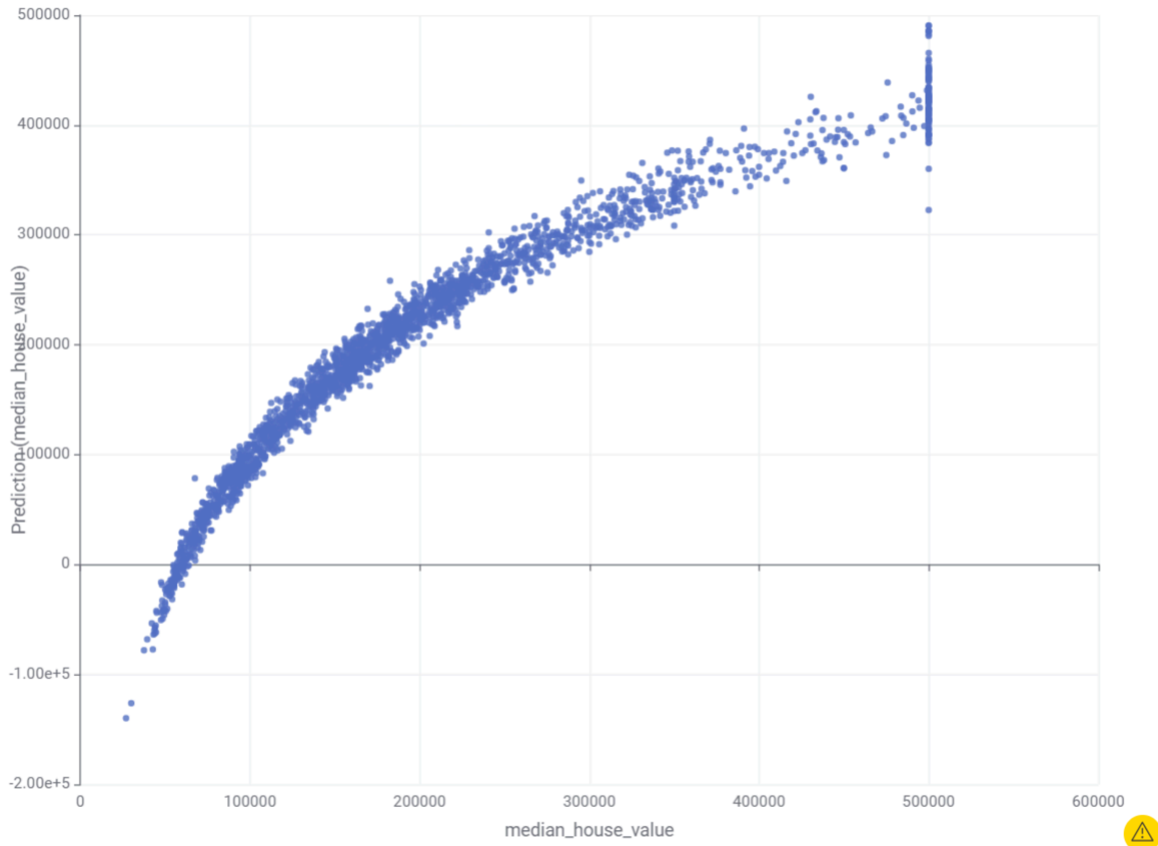
An R^2 value of 0.914 indicates that approximately 91.4% of the variance in median housing values can be explained by the predictor variables included in the model. This suggests that the selected demographic and geographic features are highly effective in explaining housing price variation across California neighborhoods.

The MAE of $25,604 means that, on average, the model's predicted housing prices differed from actual housing prices by about $26,600. Given the wide range of housing prices in the dataset, this represents a relatively accurate prediction margin.

The MAPE of 19% indicates that the model's predictions were, on average, within 19% of the true housing value. This level of percentage error demonstrates that the model provides reasonably accurate pricing estimates while still leaving room for improvement.

Compared to a basic linear model built before engineering and feature selection, the final model represented approximately 48% improvement in predictive performance, showing that preprocessing and feature engineering substantially improved model quality.

These performance metrics indicate that the final model provides strong predictive capability and captures the majority of variation in California housing prices, demonstrating that the selected features are highly informative for housing value prediction.



Predicted vs. Actual Median Housing Values from the Final Linear Regression Model. Points clustering near the diagonal indicate strong predictive accuracy.

Discussion

Overall, the data suggests that California housing prices can be predicted with relatively high accuracy using a combination of demographic and geographic variables. The strong model performance indicates that housing value in California is heavily influenced by measurable neighborhood characteristics, particularly economic and location-based features.

In our opinion, the most important takeaway from the dataset is how strongly location and income dominate housing prices. Geographic regions consistently showed clear pricing differences during exploratory analysis, and median income proved to be one of the strongest numerical predictors in the regression model. This aligns with real-world expectations, as desirable coastal or bay-adjacent locations and wealthier neighborhoods typically command higher housing prices.

Another interesting observation is that raw housing size variables such as total rooms and bedrooms were less useful independently than expected due to multicollinearity and redundancy. This suggests that broader neighborhood-level economic and demographic indicators may matter more for predicting median housing prices than individual housing stock characteristics alone.

However, while the model performed well, the data also reveals that housing prices are influenced by factors beyond those captured in the dataset. Real estate markets are complex and are shaped by many economic, political, and social influences that cannot be fully represented through demographic and geographic variables alone.

Overall, we believe the dataset provided a strong foundation for predictive modeling and successfully demonstrated that machine learning techniques can produce highly accurate housing price predictions. The results support the conclusion that both economic and geographic factors play major roles in shaping California's housing market.

Limitations/Challenges

Overall, the dataset had several limitations for accurately predicting the housing value. The scope of the actual problem is limited not only by the amount of attributes in each column, but also by the actual values of the columns. One example is that the housing value was capped at $500,000. This created an inaccurate representation of houses that were valued higher, causing a massive spike in the dataset.

Some missing values that could wildly affect each house's value, or reveal more information would be interest rates, employment opportunities, homeowner turnover, and inflation rate. The timeframe could be an issue, but inflation rate from the past could be used to predict values in the future as well. With more values, it may take longer for the pipeline to process and create models, however there would be more accuracy with certain exact areas. The current dataset only had broad area types and if places were close to the sea.

Total rooms and bedrooms was also an interesting datapoint. It actually showed how many bedrooms and rooms were in the neighborhood, not the individual home. This could lead to overprediction in value if the home had fewer bedrooms than the average home in its neighborhood.

Conclusion

Overall we were able to create an accurate predictor for the housing value based on demographic and geographic data. Median income and location were the most important predictors for the value of the homes. Housing markets are inherently complex, and this was made clear to us due to the limitations of the dataset given. More granular housing data could significantly improve the accuracy of our predictive model.

The project shows how using machine learning and data mining pipeline techniques can uncover patterns in housing data that can be used to predict the values of homes. Data selection is very important, especially when dealing with complex real-world data.